

- There is an important connection between invertible matrices and row operations that leads to a method for computing inverses.
- We shall see that an invertible matrix A is row equivalent to an identity matrix, and we can find A^{-1} by watching the row reduction of A to I . $\square$

Elementary Matrices

An elementary matrix is one that is obtained by performing a single elementary row operation on an identity matrix.

Example 5: Let $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Compute $E_1 A$, $E_2 A$ and $E_3 A$, and describe how these products can be obtained by elementary row operations on A .

Solution: $E_1 A = \begin{bmatrix} E_{11} & E_{12} & E_{13} \end{bmatrix}$ (By defn. of matrix multiplication)

$$E_1 a_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ d \\ g \end{bmatrix} = \begin{bmatrix} 1 \cdot a + 0 \cdot d + 0 \cdot g \\ 0 \cdot a + 1 \cdot d + 0 \cdot g \\ (-4) \cdot a + 0 \cdot d + 1 \cdot g \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$= \begin{bmatrix} a \\ d \\ g - 4a \end{bmatrix}$$

$$E_2 a_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} b \\ e \\ h \end{bmatrix} = \begin{bmatrix} 0 \cdot b + 1 \cdot e + 0 \cdot h \\ 1 \cdot b + 0 \cdot e + 0 \cdot h \\ 0 \cdot b + 0 \cdot e + 1 \cdot h \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$= \begin{bmatrix} e \\ b \\ h \end{bmatrix}$$

$$E_3 a_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \begin{bmatrix} b \\ e \\ h \end{bmatrix} = \begin{bmatrix} b \\ e \\ h - 4b \end{bmatrix} \quad (\text{Similar calculation as } E_1 a_1)$$

$$E_1 a_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \begin{bmatrix} c \\ f \\ i \end{bmatrix} = \begin{bmatrix} c \\ f \\ i - 4c \end{bmatrix} \quad (\text{Similar calculation as } E_1 a_1)$$